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Roll No

AU-802 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Vehicle Dynamics
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain Newtonian and Lagrangian formulation methods of investigation, with neat sketches wherever necessary.
2. Explain the various static and Rolling properties of a tyre. Also explain rolling resistance.
3. With the help of a neat sketch describe the construction of tires. How construction of tubeless tyre is different from tubed tyres?
4. Explain the effect of mass distribution and engine location on stability and handling of a 3-wheeler and 4-wheeler separately.
5. Explain Multi-body system approach used in the study of Vehicle Dynamics.

OR

What are tire lugs and voids? Explain how tires tread pattern effects tire performance.

6. Draw the characteristic curve of an automobile. Elaborate the different transmission characteristics.
7. a) What is steering geometry? Explain.
b) Explain various transient response characteristics.
8. Write a short note on the following:
a) Suspension isolation
b) Suspension damping
